

Business Questions

Revenue & Growth

1. What is the best performing product category? (by revenue?)

Answer:

Health and Beauty is the top performing product category based on gross revenue. Revenue was calculated by summing the price of all order line items associated with each product category. Health and Beauty's gross revenue totaled \$1,258,681.34. The next highest performing product category was watch gifts which totaled \$1,205,005.68. It is a surprising find that watches would be so close in revenue to something like Health and Beauty which probably has significantly more purchases. With that being said, watches cost significantly more on average making up for the difference in order amounts. I made a deliberate choice to use `order_items.price` rather than `payment_value` for category analysis because I wanted to isolate product revenue from ancillary fees. For total business revenue calculations I used the payments table instead.

2. What is the total monthly revenue by product category?

Answer:

Health and Beauty November 2017 revenue was 97% above the monthly average, suggesting a strong Black Friday driven demand spike. This pattern was consistent across all top 5 revenue categories indicating a business wide seasonality effect rather than a category specific anomaly.

3. What is the month over month revenue growth rate?

Answer:

From February 2017 through August 2018 the overall business averaged 9.17% month over month revenue growth. Four months exceeded 25% growth: March 2017 (51.37%), May 2017 (40.6%), November 2017 (52.1%), and January 2018 (27.71%). The November spike aligns directly with Question 2's finding that November is the highest revenue month in the dataset, consistent with Black Friday driven demand.

To compare growth across years on equal footing, I isolated the months available in both years (March through August) and calculated the average growth rate for each period independently. The 2017 average was 12.32% compared to 0.6% in 2018, meaning 2017 grew at roughly 25 times the rate of 2018 during the same seasonal window. This suggests that the

business was in a hyper growth phase in 2017 as it scaled its customer base and market presence. The significantly lower but stable 2018 rate is more consistent with a maturing business and should be treated as the more reliable baseline for future growth expectations.

Analyzing the top performing category separately, Health & Beauty averaged 13.9% month over month growth over the same period, outpacing the overall business average. Health & Beauty consistently mirrors the business's peak growth months and frequently exceeds the overall average during those periods. Given that it is the number one revenue generating category, Health & Beauty appears to function as a leading indicator for overall business performance. When this category spikes, the broader business follows. This pattern has meaningful implications for inventory planning, marketing investment, and demand forecasting.

4. What is the average order value overall and how does it vary by category?

Answer:

The average order value for the entire business is \$144.78. The top 5 categories in average order value were computers (\$1098.34), small_appliances_home_oven_and_coffee (\$624.29), home_appliances_2 (\$476.12), agro_industry_and_commerce (\$342.12), musical_instruments (\$281.62). The fact that not one of these top 5 AOV categories appears in the top 10 for total revenue illustrates that high transaction value alone doesn't drive revenue. Volume is the multiplier that determines which categories actually move the needle. Health and Beauty has a higher AOV than average (\$130.16) and very high order volume which continues to prove the point that this category is the driving force of the business.

Customer Behavior

5. Which product category received the most 1-star reviews, and what percentage of that category's total reviews were 1-star?

Answer:

Office furniture had the highest 1-star review rate at 21% among product categories with at least 500 reviews. This is a notable gap above the next closest category. Three of the top 5 categories with the highest 1-star rates are furniture-related: office furniture (21%), furniture decor (15.1%), and furniture living room. This finding suggests a pattern worth investigating, whether that points to shipping damage, assembly issues, or product quality.

Bed, bath, table is arguably the most actionable finding. It came in fourth with a 14.5% 1-star rate but has the highest total review volume in the top 5, meaning the raw number of dissatisfied customers is significant. Combined with the fact that bed bath table is a top 5

revenue category from my earlier analysis, this is the category where addressing complaints could have the greatest business impact. Understanding why customers are leaving 1-star reviews here should be a priority.

6. What percentage of customers make more than one purchase?

Answer:

Only 3.12% of customers made more than one purchase, meaning nearly 97% of the customer base bought once and never returned. This indicates the business is almost entirely acquisition-driven. The growth of this business depends on continuously finding new customers rather than retaining existing ones. That recontextualizes prior findings: the revenue growth identified in Q3 and Health & Beauty's dominance in Q2 are likely being sustained by new customer volume, not loyalty.

7. Which states or regions generate the most orders and revenue?

Answer:

São Paulo (47,449 orders, \$5,202,955), Rio de Janeiro (14,579 orders, \$1,824,093), Minas Gerais (13,129 orders, \$1,585,308), Rio Grande do Sul (6,235 orders, \$750,304), and Paraná (5,740 orders, \$683,084) are the top five states by both order volume and revenue, and their rankings are identical across both metrics.

São Paulo is in a league of its own and generates more than 3x the orders and revenue of second-place Rio de Janeiro, which reflects its status as Brazil's largest and most economically dominant state. What's surprising is that every top 5 state has a revenue per order below the overall dataset average of \$144.78. São Paulo (\$109.65), Rio de Janeiro (\$125.12), Minas Gerais (\$120.98), Rio Grande do Sul (\$120.34), Paraná (\$118.18). São Paulo has the lowest of the group despite driving the most volume. This suggests the overall average is being pulled upward by smaller, lower-volume states where customers tend to spend more per order. This is a pattern worth investigating further.

8. What is the average customer review score by product category?

Answer:

The top 5 categories by average review score (minimum 500 reviews) are books_general_interest (4.45), luggage_accessories (4.32), stationery (4.19), pet_shop (4.19), and home_appliances (4.17). The bottom 5 are office_furniture (3.49), furniture_living_room (3.90), furniture_decor (3.90), bed_bath_table (3.90), and computers_accessories (3.93).

Most categories fall within a tight band between 3.90 and 4.20, suggesting satisfaction is fairly consistent across the business. The exception is office_furniture, which sits 0.41 points below the next lowest category. This is a meaningful gap that points to a genuine quality or expectation problem, not just normal variation.

Two bottom-tier categories connect directly to Q5 findings: office_furniture had the highest 1-star review rate at 21%, and bed_bath_table was flagged as the most actionable concern due to its high review volume and top-5 revenue status. Both metrics now tell the same story for these categories.

Operational Performance

9. How long does it take for an order to be delivered after it has been placed?

Answer:

The overall average delivery time is 12.05 days from order placement to customer receipt and delivery. Delivery times vary significantly across categories, arts_and_craftsmanship is the fastest at 5.72 days, while office_furniture is the slowest at 20.84 days, a gap of over 15 days between the two extremes. Office_furniture is nearly 5 days slower than the second slowest category, christmas_supplies at 15.74 days, making it a genuine outlier rather than just the tail end of a normal distribution. The office_furniture finding becomes more meaningful alongside Q5 and Q8: this category already had the highest 1-star review rate and the lowest average review score in the dataset. Delivery time is a plausible contributor to that dissatisfaction. Customers waiting over three weeks for furniture are likely holding it to a higher standard than a craft supply arriving in under 6 days.

10. What products have the longest average delivery time?

Answer:

A product in the auto category takes the longest to deliver on average at 35.47 days. Looking at the bigger picture of this specific category, it can be determined that this product is an extreme outlier and does not reflect the average delivery time of the auto category as a whole, which is 12.23 days. This is due to the fact that this specific product only had 35 deliveries while the auto category had over 4,000 deliveries as a whole, so the long wait time on this specific product does not significantly impact the auto category average. This finding is still worth noting, however, as the slowness appears to be consistent and real rather than a data artifact.

The most valuable finding here is that 12 of the top 25 slowest individual products belong to the office_furniture category. This is the same category that already had the slowest average delivery time in the dataset at 20.84 days in Q9, the highest 1-star review rate in Q5, and the lowest average review score in Q8. The product-level view confirms what the category-level analysis already suggested: office_furniture has a systemic operational problem that is directly showing up in customer dissatisfaction. This is not noise, it is a pattern that has now surfaced consistently across four separate questions.

11. What percentage of orders are delivered late relative to the estimated delivery date?

Answer:

The overall late delivery rate is 8.11%, which actually compares favorably against industry benchmarks. The industry average for on-time delivery sits between 85-90%, implying a typical late rate of 10-15%, meaning Olist is performing closer to the top-quartile threshold than the middle of the pack.

An interesting finding from the category-level breakdown is that office_furniture does not appear in the top 10 highest late-rate categories despite being the slowest category by absolute delivery time in Q9. This suggests that while office_furniture takes significantly longer to deliver than any other category, it tends to set realistic customer expectations upfront. Orders are slow, but they arrive roughly when promised. This is worth noting, though it does not diminish the operational problems identified across Q5, Q8, Q9, and Q10. A category that pads its estimates to avoid looking late is solving a metrics problem, not a fulfillment problem.

The five categories with the highest late delivery rates are home_comfort_2 (16.67%), furniture_mattress_and_upholstery (13.51%), audio (12.71%), fashion_underwear_beach (12.6%), and christmas_supplies (12.0%). Home_comfort_2 carries a naming anomaly suggesting a data quality issue and should be interpreted cautiously. The remaining four are more

meaningful. Furniture and large item categories are a known challenge industry-wide, with furniture and appliances achieving only a 23% two-day delivery rate due to size constraints, so furniture_mattress_and_upholstery's appearance near the top is consistent with broader industry patterns. Audio and fashion categories appearing alongside furniture are more notable and potentially more actionable, as those product types do not carry the same inherent logistical constraints.

12. What payment methods do customers prefer and does payment type correlate with order value?

Answer:

Credit cards are by far the most common payment type, accounting for 76.94% of all orders. It also has the highest average order value at \$163.32. This is consistent with Brazilian consumer culture where credit card installment plans are deeply embedded, allowing customers to spread large purchases over time, which likely explains why credit card correlates with higher order values as well.

The second most common payment type is boleto at 19.9% with an average order value of \$145.03. Together credit card and boleto account for over 95% of all orders, and this dominance is consistent across every product category in the dataset.

Voucher is worth calling out specifically. It accounts for more than twice the orders of debit card (3.89% vs 1.54%), but has an average order value of just \$65.70. That is less than half of credit card payments and well below every other payment type. This suggests vouchers are used for small or discounted purchases rather than full price orders, not as a primary payment method for regular shopping.

Seller & Product Performance

13. Who are the top 10 sellers by revenue and how do their average review scores compare?

Answer:

The top seller leads both in revenue (\$228,071) and customer satisfaction with a 4.12 average review score, suggesting strong all-around performance. Most of the top 10 sellers

cluster near the platform's overall average review score of around 4.0, which indicates that high revenue and solid satisfaction tend to go hand in hand at the top of the seller base.

The most notable exception is the 5th-highest revenue seller, who has an average review score of just 3.35. That average score is lower than office_furniture's 3.73, which was the lowest-scoring product category identified in previous analysis. This seller is generating nearly \$188,000 in revenue despite significant customer dissatisfaction. Understanding what is driving those low scores represents a high-priority opportunity, since resolving the underlying issues could protect and potentially grow revenue from an already high-performing seller.

14. Which product categories have the highest cancellation or return rates?

Answer:

Cancellation rates are very low across the board, with musical_instruments posting the highest rate among categories with 500 or more orders at just 1.75%. The top three categories by cancellation rate which are musical_instruments, books_general_interest, and small_appliances all have under 650 total orders, meaning their impact on the business is limited. By contrast, high-volume categories like sports_leisure (7,720 orders) sit near the bottom of the cancellation ranking at just 0.66%, suggesting that the categories driving the most business are also the most stable.

Office_furniture, which surfaced as a problem category across delivery time, review scores, and product-level analysis, has one of the lowest cancellation rates on the platform. Customers appear committed to completing furniture purchases even when the experience is poor, which means the risk for that category is not lost revenue from cancellations but damage to reputation through low review scores. Overall, cancellation is not a primary concern for this business. The more meaningful operational risks identified throughout this project lie in delivery performance and customer satisfaction.

15. What percentage of total revenue comes from the top 10% of sellers?

Answer:

The top 10% of sellers (309 out of 3,095 total) account for 67.49% of overall platform revenue. To put that in perspective, the remaining 90% of sellers share just 32.51% of revenue between them, meaning the vast majority of the seller base contributes relatively little to the bottom line.

This level of revenue concentration has clear strategic implications. Losing even a small number of top sellers could have a meaningful and immediate impact on platform revenue. The recommendation would be to allocate dedicated resources and attention to this group, whether through account management, preferential support, or incentive programs. This would help them grow and remain on the platform. Retaining top sellers is likely one of the highest-leverage investments this business can make.

Python Analysis Answers

Question 6: What percentage of customers make more than one purchase?

Answer:

SQL analysis revealed that only 3.12% of customers placed more than one order. Python pandas analysis provided deeper insight into the behavior of these repeat customers. Of the 3,045 customers who placed multiple orders, 93% placed exactly two orders, suggesting these were not truly loyal customers but rather occasional repeat buyers. Only 7% of repeat customers placed three or more orders, indicating a significant retention cliff between first and second repeat purchases. Combined with the November seasonality finding from Question 2, this suggests Olist is heavily dependent on seasonal demand to acquire new customers rather than retaining existing ones which is a pattern that is not sustainable at scale.

Question 9: How long does it take for an order to be delivered after it has been placed?

Answer:

Delivery time analysis revealed that the average order takes 12 days to reach a customer from the time of purchase, with a median of 10 days. However a standard deviation of 9.55 days indicates highly inconsistent delivery performance, suggesting that while some customers receive orders within a week, many others experience significantly longer wait times. 301 orders exceeding 60 days were identified and excluded as outliers, representing 0.31% of all delivered orders. These extreme cases likely represent lost, damaged, or disputed shipments rather than standard operational delays.

Regional analysis revealed a strong correlation between state infrastructure and delivery performance. São Paulo, Brazil's largest economic hub and the state with the highest order volume at 40,431 orders, had the fastest average delivery time at 8.19 days, likely reflecting seller concentration in the region. Conversely Amazonas recorded the slowest average delivery

time at 25.21 days, a finding that is consistent with the logistical complexity of the region's geography and limited infrastructure.

Question 11: What percentage of orders are delivered late relative to the estimated delivery date?

Answer:

SQL analysis identified a 7.87% late delivery rate across all orders. Python pandas analysis revealed deeper insights into both the magnitude of these delays and their impact on customer satisfaction. The average late order arrives 8.87 days past the promised delivery date, with 25% of late orders exceeding 11 days past the estimated delivery date.

Most significantly, a merge of delivery performance data with customer review scores revealed a direct correlation between late deliveries and customer dissatisfaction. On time deliveries received an average review score of 4.21 out of 5, while late deliveries received an average of just 2.57, a 39% drop in customer satisfaction directly attributable to operational delays.

Revenue at risk analysis identified R\$1,357,314.96 which is 8.44% of total revenue, as being associated with late orders and dissatisfied customers. Given that only 3.12% of customers return for a second purchase, customers who experienced late deliveries are highly unlikely to be retained, compounding the long term revenue impact.

Question 13: Who are the top 10 sellers by revenue and how do their average review scores compare?

Answer:

SQL analysis identified the top 10 sellers by revenue and their corresponding average review scores. Two sellers were flagged for further investigation based on below average review scores of 3.35 and 3.80 respectively.

Python pandas analysis revealed that surface level averages were misleading in both cases. The seller with a 3.35 average generates 99% of their revenue from watches gifts, where their actual average review score is 4.35. Their poor overall average was caused entirely by 3 outlier orders in secondary categories representing less than 1% of their business. This finding demonstrates how aggregate metrics without categorical context can misrepresent seller performance and lead to incorrect business decisions.

The second seller flagged at 3.80 was similarly vindicated upon deeper analysis. Investigation revealed that null review values during data merging artificially suppressed their true average of 4.08. Category level analysis confirmed strong performance across both product lines, computers at 4.19 and telephony at 4.01 across a combined 410 orders.

Question 15: What percentage of total revenue comes from the top 10% of sellers?

Answer:

SQL analysis revealed that the top 10% of sellers account for 67.49% of total revenue. Python pandas analysis provided a deeper look at revenue distribution across the entire seller base, revealing a far more nuanced and strategically significant picture. The most striking finding was that the top 1% of sellers (30 individuals) generate 25.67% of total platform revenue. The average top 10% seller generates R\$29,688 compared to R\$282 for the average bottom 50% seller, a performance gap of 105 times. This level of concentration at the very top represents a genuine business risk. Losing even 2-3 of these top sellers would produce an immediate and negative measurable revenue impact.

An equally significant finding is that the bottom 50% of sellers, (1,547 sellers) contribute only 3.22% of total revenue collectively, with the lowest performing seller generating just R\$3.50 in total transactions. Rather than removing these sellers which would reduce product variety and platform competitiveness, a more strategic approach would be to investigate why these sellers are underperforming and implement a structured development program to help them grow.